so KiCad pcb design and FEntwumS FPGA logic gates and what is the free open tool for SoC SiP MCM PoP ASIC SoM CoM for circuits design

Researched open-source tools for SoC, ASIC, and packaging

The Full Stack: Open-Source Tools by Integration Level

The open-source landscape has a **critical cliff** at the packaging level. Everything below the die is well-served; everything at and above the package is almost entirely proprietary.

What exists (by domain)

Domain	Open-Source Tool	License
FPGA logic	FEntwumS (Yosys + nextpnr + GHDL + liteX)	ISC / EUPL / (
PCB / board	KiCad 9	GPL-2.0+
SoC / ASIC (RTL→GDSII)	OpenLane / LibreLane / SiliconCompiler	Apache-2.0
ASIC physical design	OpenROAD	BSD-3
RTL synthesis	Yosys	ISC
GDSII layout/view	KLayout	EUPL-1.2
Analog / mixed-signal ASIC	OpenFASoC (Xschem + ngspice + Magic)	GPL / EUPL
SoM / CoM	KiCad (same as PCB)	GPL-2.0+

The critical gap: SiP / MCM / PoP / 3D IC packaging

There is no open-source tool for this domain. The entire SiP/MCM/PoP design flow is locked behind proprietary licenses:

Function	Proprietary tools	Open-source equiv
Die-on-substrate placement	Cadence Allegro X APD, Synopsys IC Compiler II (3DIC)	**None**
BGA ball assignment / via stitching	Cadence, Zuken CR-8000	**None**
2.5D/3D interposer design	Synopsys, Cadence	**None**
Package-level SI/PI analysis	Ansys SIwave/Icepak, Siemens Calibre 3DSTACK	MEEP (partial — FD no package-specific DRC/routing)
Thermal (multi-die)	Ansys Icepak	Elmer FEM / OpenF (general-purpose, n specific)
Package substrate routing	Cadence, Zuken	**KiCad** (hackable — substrate as a very l count PCB, but no D package rules, no di constraints)

The closest you can get with open tools:

- **KiCad** for the substrate layout (it supports up to 60 layers, which covers most organic substrates)
- **MEEP** for electromagnetic simulation of the package structure
- **Elmer FEM / OpenFOAM** for thermal analysis
- **KLayout** for viewing the GDSII of individual dies before integration

But there is **no integrated flow** that takes multiple GDSII dies, places them on a substrate, assigns balls, stitches vias, and runs package-level SI/PI/thermal signoff. This is a **genuine gap** with no active open-source project I can find working to fill it.

SoM / CoM: no special tool needed

SoM and CoM design is **just PCB design in a constrained form factor**. The workflow is:

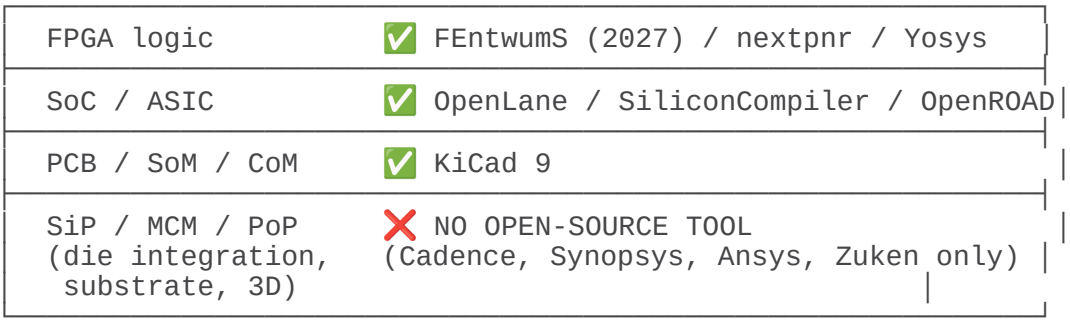
1. Choose a SoC (Raspberry Pi CM5, Toradex Verdin, Microchip SAMA5D27, etc.)
2. Design the carrier board in **KiCad** (power delivery, memory, connectivity, connectors)
3. Follow the vendor's pinout/mechanical spec

Open-source SoM/CoM projects that publish full KiCad source:

- **Raspberry Pi CM4 / CM5** — official KiCad files on GitHub
- **Microchip SAMA5D27 SOM** — open schematics, Gerber, BOM
- **OSM (Open Standard Module)** — open hardware + software standard (postage-stamp form factor, LGA, no connector)
- **Toradex** — reference carrier board schematics (not full KiCad, but detailed enough)

No dedicated "SoM design tool" exists or is needed — KiCad + the vendor's mechanical spec is the standard workflow.

Summary: where the open-source cliff is



The SiP/MCM/PoP layer is where the open-source ecosystem **completely breaks**. If you need to design a multi-die package (e.g., stacking a logic die with an HBM die on an interposer, or a PoP with DRAM over a base die), you are currently locked into **$50K–$200K/year proprietary tool licenses** with no viable open alternative. This is the single largest gap in the entire open hardware design stack.